Experiment 14.1.1: Complex Number Addition using Class

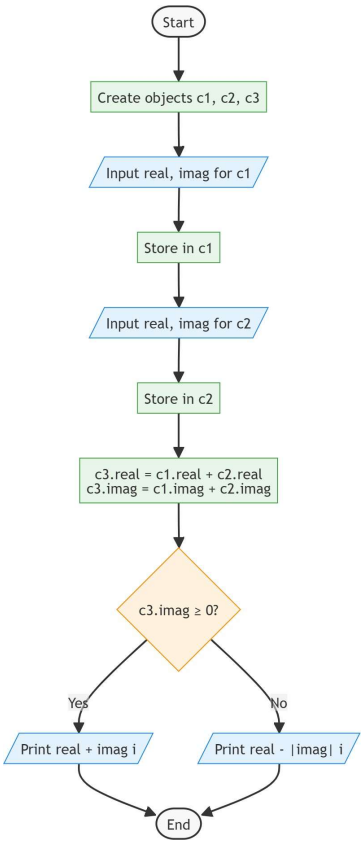
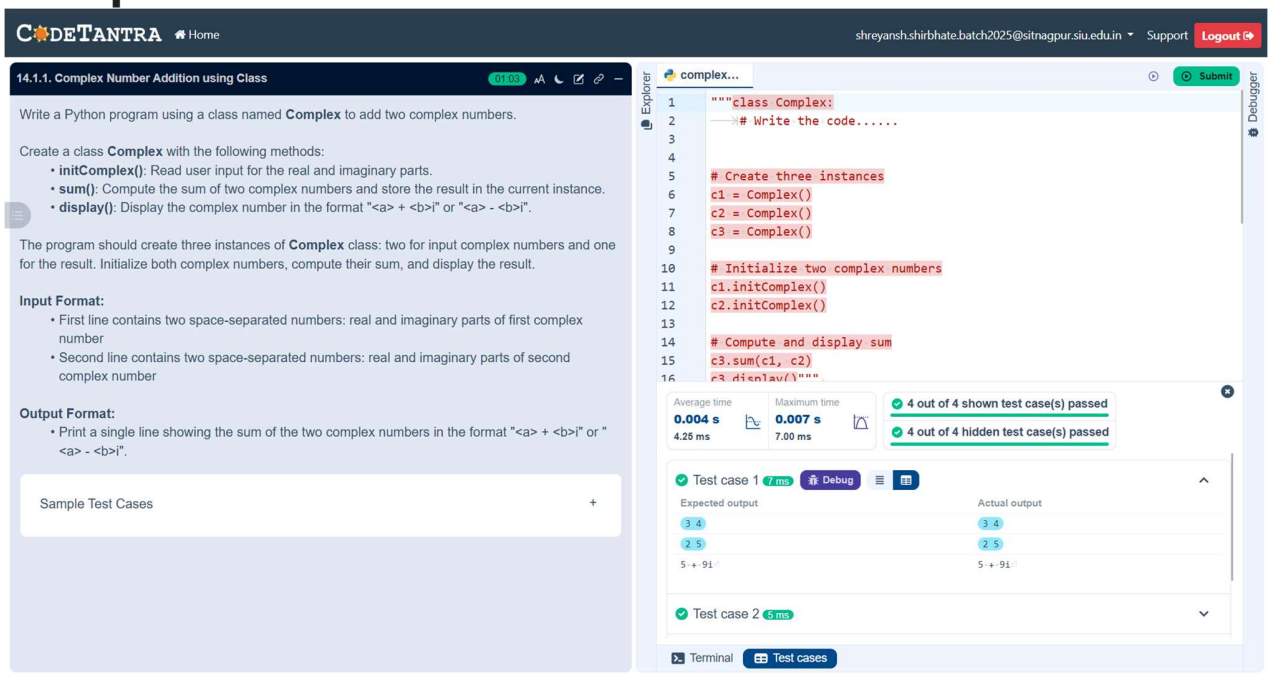
1. Aim

To design and implement a Python program using a class named **Complex** to add two complex numbers. The program involves using methods to initialize values from user input, compute the sum of two complex instances, and display the result in a standard complex number format ($a + bi$ or $a - bi$).

2. Pseudocode

1. **START**
2. **DEFINE** a class Complex:
   - **METHOD** initComplex(self):
     - **READ** two space-separated integers (real and imaginary parts) and **STORE** them in self.real and self.imag.
   - **METHOD** sum(self, c1, c2):
     - **SET** self.real = c1.real + c2.real.
     - **SET** self.imag = c1.imag + c2.imag.
   - **METHOD** display(self):
     - **IF** self.imag >= 0:
       - **PRINT** the complex number in the format: real + imag i.
     - **ELSE**:
       - **PRINT** the complex number in the format: real - abs(imag) i.
3. **MAIN PROGRAM**:
   - **CREATE** three instances of the Complex class: c1, c2, and c3.
   - **CALL** c1.initComplex() to read the first complex number.
   - **CALL** c2.initComplex() to read the second complex number.
   - **CALL** c3.sum(c1, c2) to compute the addition.
   - **CALL** c3.display() to output the result.
4. **END**

3. Flowchart



4. Experiment Screenshot